$L_1 := 1\ m$ $L_2 := 1\ m$ $L_3 := 1\ m$ $L_4 := 1\ m$ $L_5 := 1\ m$

$L_6 := 1\ m$ $L_7 := 1\ m$ $L_8 := 1\ m$ $L_9 := 0\ m$ $L_{10} := 0\ m$

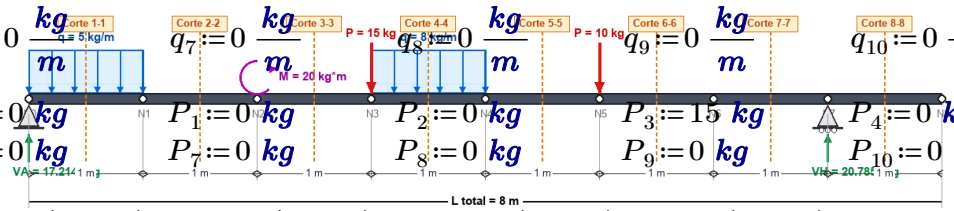
$x_0 := 0.0\ m$ $x_1 := L_1$ $x_2 := x_1 + L_2$ $x_3 := x_2 + L_3$ $x_4 := x_3 + L_4$

$x_5 := x_4 + L_5$ $x_6 := x_5 + L_6$ $x_7 := x_6 + L_7$ $x_8 := x_7 + L_8$ $x_9 := x_8 + L_9$ $x_{10} := x_9 + L_{10}$

$L_{tot} := x_{10}$

$q_1 := 5\ \frac{kg}{m}$ $q_2 := 0\ \frac{kg}{m}$ $q_3 := 0\ \frac{kg}{m}$ $q_4 := 8\ \frac{kg}{m}$ $q_5 := 0\ \frac{kg}{m}$

$q_6 := 0\ \frac{kg}{m}$ $q_7 := 0\ \frac{kg}{m}$ $q_8 := 0\ \frac{kg}{m}$ $q_9 := 0\ \frac{kg}{m}$ $q_{10} := 0\ \frac{kg}{m}$



$P_0 := 0\ kg$ $P_1 := 0\ kg$ $P_2 := 0\ kg$ $P_3 := 15\ kg$ $P_4 := 0\ kg$ $P_5 := 10\ kg$

$P_6 := 0\ kg$ $P_7 := 0\ kg$ $P_8 := 0\ kg$ $P_9 := 0\ kg$ $P_{10} := 0\ kg$

$M_0 := 0\ (kg \cdot m)$ $M_1 := 0\ (kg \cdot m)$ $M_2 := 20\ (kg \cdot m)$ $M_3 := 0\ (kg \cdot m)$ $M_4 := 0\ (kg \cdot m)$ $M_5 := 0\ (kg \cdot m)$

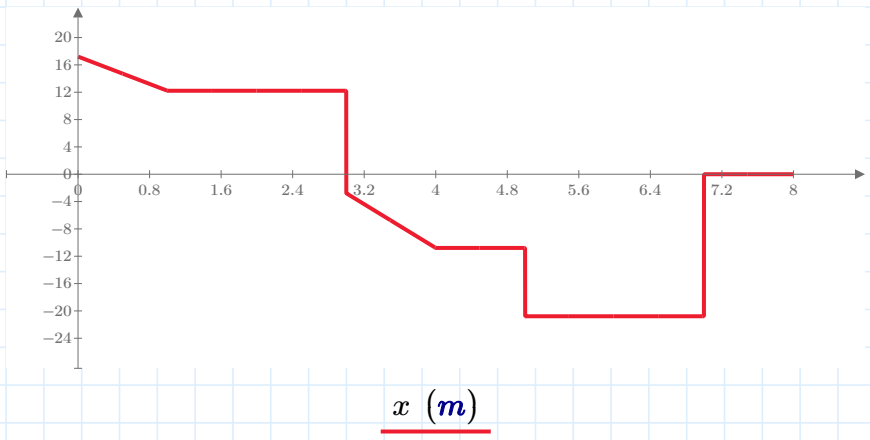
$M_6 := 0\ (kg \cdot m)$ $M_7 := 0\ (kg \cdot m)$ $M_8 := 0\ (kg \cdot m)$ $M_9 := 0\ (kg \cdot m)$ $M_{10} := 0\ (kg \cdot m)$

$V_D := 20.7857\ kg$ $V_A := 17.2143\ kg$

$$M(x) := V_A \cdot x + (x \geq x_7) \cdot V_D \cdot (x - x_7) - (x_1 > x > x_0) \cdot q_1 \cdot \frac{x^2}{2} - (x \geq x_1) \cdot q_1 \cdot L_1 \cdot \left(x - \frac{L_1}{2}\right) - (x_4 > x > x_3) \cdot q_4 \cdot \frac{(x - x_3)^2}{2} - (x \geq x_4) \cdot q_4 \cdot L_4 \cdot \left(x - \left(x_3 + \frac{L_4}{2}\right)\right) + ($$

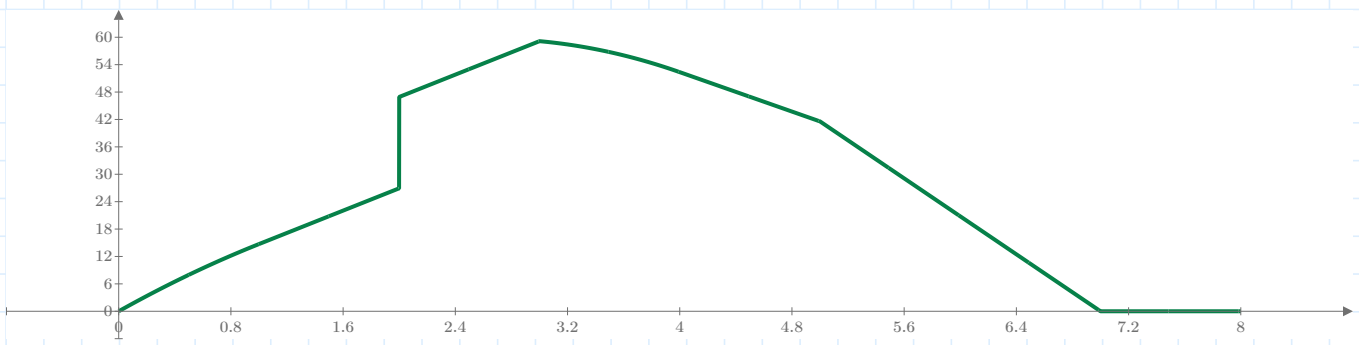
$$Q(x) := V_A + (x \geq x_7) \cdot V_D - (x_1 > x > x_0) \cdot q_1 \cdot x - (x \geq x_1) \cdot q_1 \cdot L_1 - (x_4 > x > x_3) \cdot q_4 \cdot (x - x_3) - (x \geq x_4) \cdot q_4 \cdot L_4 - (x \geq x_3) \cdot P_3 - (x \geq x_5) \cdot P_5$$

$x := 0.0\ m, 0.001\ m .. L_{tot}$



$Q(x)\ (kg)$

$x\ (m)$



$M(x)\ (kg \cdot m)$

$x\ (m)$